

LC 2492: Min Score of a path between 2 cities

UnBxd

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Key Observations

- weighted graph, but no need to run Dijkstras
- we dont need to minimize sum of edges, we only need minimum edge on the path
- so start DFS from city 1, traverse whole connected component
- everytime we see edge: **`ans = min(ans, w)`**

Algorithm

1. build adjacency list
2. DFS from city 1
3. maintain `ans = INT_MAX`
4. `ans = min(ans, weight)`

C++ Implementation

```

class Solution {
public:
    int minScore(int n, vector<vector<int>>& roads) {

        int ans=INT_MAX;
        vector<vector<pair<int,int>>> adj(n+1);
        vector<int> seen(n+1,0);

        for(auto &it:roads){
            int u=it[0];
            int v=it[1];
            int w=it[2];

            adj[u].push_back({v,w});
            adj[v].push_back({u,w});
        }

        //will store cities in queue
        //distance se koi matlab nhi
        queue<int> q;
        q.push(1);
        seen[1]=1;

        while(!q.empty()){
            int currcit=q.front();
            q.pop();

            for(auto &it:adj[currcit]){
                int nextcit=it.first;
                int weight=it.second;

                ans=min(ans,weight); //already dekh bhi rkha ho to koi baat ni
                                     //kal ko bhatka hua raat ko
                                     //question is basically to find
                                     //minimum edge cost in whole
                                     //kyunki there is a guarenteed path
                                     //to ghum phir ke bhi ja skte
                                     //seen same city multiple times

                if(!seen[nextcit]){
                    seen[nextcit]=1;
                    q.push(nextcit);
                }
            }
        }
    }
}

```

```
        }  
    }  
  
    return ans;  
}  
};
```